PADHAKU
Exploration · Class 9 Science



CHAPTER 3

Tissues in Action

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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🤔 Think It Over

1 **How is the study of cells and tissues significant for understanding the life processes and human welfare?**

💡 SAMPLE ANSWER

- All life processes (growth, movement, nutrition, repair) are done by cells and tissues, so studying them explains how the body works.
- It helps in human welfare: understanding diseases, healing wounds, tissue culture to grow many plants, better crops and new medical treatments.

2 **How are tissues in plants and animals different, and why?**

✔ ANSWER

- Plants are fixed in one place and need support, so many plant tissues have cell walls and some are dead (e.g. sclerenchyma, xylem). Plants have meristems that keep dividing at certain places, so they grow throughout life.
- Animals move to find food, so they have flexible cells and special tissues like muscle and nervous tissue; most animal tissues are living.

So the difference is due to their different ways of life: fixed vs moving, making food vs eating food.

3 **How is the division of labour at various levels of organisation in multicellular organisms correlated with their structure and function?**

✔ ANSWER

Cells → tissues → organs → organ systems → organism. At each level, a group is shaped for one job: e.g. long muscle fibres contract for movement, long neurons carry messages, tube-like xylem carries water.

Because each part does a special job, the body works efficiently and can carry out complex life processes.

Activity 3.1 · 3.2.1 Apical meristem

Activity 3.1 **Grow onion bulbs in Jar A and Jar B. On day 3 cut about 1 cm of the root tips in Jar B. Measure root lengths for 7 days (Table 3.1).**

What trend do you observe? Are your observations similar to the graph (Fig. 3.2)? What do you infer?

✔ ANSWER

Observation: Roots in Jar A keep growing longer every day. Roots in Jar B grow until day 3, but after the tips are cut they almost stop growing.

Yes, this matches the graph in Fig. 3.2.

Conclusion: Roots grow in length only from their tips, because the tips have actively dividing cells (apical meristem).

? **Why do you think that the cells of meristematic tissues lack vacuoles?**

✔ ANSWER

Meristematic cells divide continuously and rapidly. They need dense cytoplasm and many organelles for this, not storage space.

A large vacuole would take up space and is used for storage, which these young dividing cells do not need. So they lack vacuoles and stay small and tightly packed.

❓ In-text questions · 3.2.4 Permanent tissues

? **Look at the T.S. of the sunflower stem (Fig. 3.7). Are all the cells similar in shape and size? How many different types of tissues can you identify? What might be the reason for different types of cells and tissues?**

✔ ANSWER

No, the cells differ in shape, size and wall thickness.

Tissues seen: epidermis (with cuticle and hairs), collenchyma, parenchyma, sclerenchyma, phloem, lateral meristem and xylem.

Reason: each tissue is specialised for a different function: protection, support, storage or conduction.

? Why does a fresh twig bend but a dry twig break? Why are seed coats hard and how do aquatic plants float?

✔ ANSWER

- A fresh twig has living collenchyma (flexible walls) and water, so it bends. A dry twig has lost water and its cells are dead and rigid, so it breaks.
- Seed coats have sclerenchyma with thick lignified walls, so they are hard.
- Aquatic plants have special parenchyma with large air spaces, which help them float.

? How does water reach the leaves of tall trees? How does food prepared in leaves reach other parts of the plant?

✔ ANSWER

Water and minerals go up from roots to leaves through **xylem**.

Food made in leaves goes to other parts through **phloem** (sieve tubes).

⭐ Pause and Ponder · 3.2.4 Permanent tissues

1 **Fibres of coconut husk are hard and brittle, whereas the leaf stalks of coriander are soft and flexible. Find out the reason.**

✔ ANSWER

Coconut husk fibres are made of **sclerenchyma**: dead cells with thick walls due to lignin, so they are hard.

Coriander leaf stalks have living parenchyma and **collenchyma** (corners thickened with flexible pectin), so they are soft and bend easily.

2 **Why is a thick cuticle on the outer wall of epidermis advantageous for a plant living in the desert but disadvantageous for a plant living underwater?**

✔ ANSWER

- Desert: water is scarce; a thick waxy cuticle reduces water loss.

- Underwater: there is no need to save water. A thick cuticle would block the exchange of gases and the absorption of water and minerals through the plant surface, which underwater plants depend on.

3 **Once water is absorbed by roots, it travels against gravity through xylem. How do the 'dead' cells of xylem work together with the living cells of leaves to keep water moving?**

✔ ANSWER

Dead xylem cells (tracheids and vessels) have no contents, so they form long, hollow, continuous tubes from roots to leaves.

Living leaf cells lose water by transpiration through stomata. This creates a **transpiration pull** that pulls the water column up through the xylem tubes.

4 **What do you think will happen if there were no stomata in the epidermis of the stem or leaves?**

✔ ANSWER

- No gas exchange: CO_2 cannot enter, so photosynthesis stops; O_2 exchange is also hampered.
- No transpiration: no transpiration pull, so water and minerals would not move up well, and the leaves would not be cooled.

The plant would become weak and could die.

❓ In-text questions · 3.3 Animal tissues

? **Blink, clench your fist, breathe deeply, touch something warm or cold. Which tissue helps you move? Which tissue enables you to sense heat or cold? Which tissue allows oxygen to enter the blood? Which tissue holds the body together so that the skin does not fall off?**

✔ ANSWER

- Move: **muscle tissue**
- Sense heat or cold: **nervous tissue**
- Oxygen into blood: **epithelial tissue** (thin, flat cells lining the lungs and blood vessels)
- Holds the body together: **connective tissue**

🧪 Activity 3.2 · 3.3.2 Connective tissue (blood)

Activity 3.2 Table 3.3: write a question for each experience.

Experience	Observation	Question
Small cut on skin	Blood oozes out, clot forms later	What causes blood to clot?
Skin infection	Area red, swollen, maybe fever	?

Exercise or running	Faster breathing, red face	?

💡 SAMPLE ANSWER

Experience	Question	Answer
Small cut	What causes blood to clot?	Platelets help blood to clot at the injury.
Skin infection	Why does the infected area become red and swollen, and why does pus form?	WBCs collect at the infected area to fight germs; this causes redness, swelling and pus.
Exercise	Why do we breathe faster and why does the face turn red?	Muscles need more oxygen, so breathing becomes faster and blood flow increases.

The red colour of blood is due to haemoglobin in RBCs.

🧪 Activity 3.3 · Connective tissues

Activity 3.3 **Perform the actions in Table 3.4 (touch elbow, press ear/nose, wiggle fingers while touching forearm, raise leg from the knee). Record experiences and identify the connective tissues.**

✔ ANSWER

Action	Experience	Tissue

Touch elbow	Hard, rigid	Bone: strength, support, protection
Press ear or nose	Soft, flexible, gets its shape back	Cartilage: flexibility, cushions bone ends
Wiggle fingers, touch forearm	Forearm moves though fingers are far	Tendon: joins muscle to bone, brings movement
Raise leg at knee	Joint stops at a limit	Ligament: joins bone to bone, gives stability

Conclusion: different connective tissues give support, flexibility, movement and stability.

🧪 Activity 3.4 · 3.4 The musculoskeletal system

Activity 3.4 **What percentage of total body weight comes from bones and muscles? Weigh yourself; use bone % (about 12–15) and muscle % to estimate bone and muscle weight; compare with classmates.**

Discuss why bone and muscle mass differ between individuals, and how they contribute to overall body weight.

💡 SAMPLE ANSWER

Example: body weight = 45 kg.

Bone weight = 45 × 12/100 to 45 × 15/100 = **5.4 kg to 6.75 kg**

Muscle weight (if 40%) = 45 × 40/100 = **18 kg**

So muscles and bones together make up about half of the body weight; muscle is the larger part.

Why they differ: age, gender, exercise and physical activity, diet (protein, calcium), heredity and body type.

Activity 3.5 · 3.4.1 The musculoskeletal system in action

Activity 3.5 **Move different body parts and fill Table 3.5: complete rotation, partial rotation, bending, other movements (Elbow: No, No, Yes is given).**

💡 SAMPLE ANSWER

Body part	Complete rotation	Partial rotation	Bending	Other movement
Elbow	No	No	Yes	Only bends and straightens
Shoulder	Yes	Yes	Yes	Arm raised sideways, forward, backward
Knee	No	No	Yes	Bends in one direction only

Neck	No	Yes	Yes	Turning side to side, up-down nodding
Fingers	No	No	Yes	Slight side spreading
Toes	No	No	Yes	Up-down
Wrist	No	Yes	Yes	Up-down, side to side, small circles

? **Joints allow movement but cannot move the bones on their own. So, what actually causes the bones to move?**

✔ ANSWER

Muscles. A muscle contracts and pulls on the bone through a tendon, and the bone moves at a joint. This is controlled by the nervous system.

⭐ **Pause and Ponder · 3.5 Types of joints**

5 **Look at Fig. 3.17 (poses of classical and folk dances of India). Can you identify which joints are involved? What type of movement does each joint allow?**

💡 SAMPLE ANSWER

Joint	Type	Movement
Shoulder and hip	Ball and socket	Arms/legs move in all directions, circular movement
Elbow and knee	Hinge	Bending and straightening in one direction
Neck (skull on backbone)	Pivot	Head turns side to side
Fingers	Hinge	Bending for hand gestures (mudras)

The skull bones (fixed joints) do not move. (check the picture in your book)

❓ In-text question · 3.6 Skeletal system

? How can such a strong rib cage move? And why does it need to move?

✔ ANSWER

The ribs are joined to the sternum in front by flexible cartilage, so the rib cage can expand and contract.

It must move for breathing: expanding increases space in the chest so air comes into the lungs; contracting pushes air out.

🤔 Think as a Scientist · Totipotency

(a) Table 3.6 (Steward's carrot phloem cells):

Light	Air	Medium	Change in fresh weight
✓	×	Solid medium + nutrients	reduced
✓	✓	Liquid medium + nutrients	20% increased
×	✓	Liquid medium + nutrients	reduced

What do you conclude about the characteristics of phloem cells of carrot?

✔ ANSWER

Phloem cells of carrot are **totipotent**: they can dedifferentiate, divide again and redifferentiate to form a whole new plant, but only under the right conditions (light, air and nutrients).

(b) In which of the three combinations would you obtain the highest and lowest biomass? What could be the possible reason(s)?

✔ ANSWER

Highest: light + air + liquid medium with nutrients (20% increase). Cells get light, oxygen and nutrients from all sides in the liquid.

Lowest: the other two (weight reduced): without air, cells get no oxygen for respiration; without light, cells cannot grow and develop properly (e.g. no photosynthesis), so they use up stored food and lose weight.

(c) **Will you get the same results if you culture animal cells instead of carrot cells?**

✔ ANSWER

No. Animal cells can divide in culture, but specialised animal cells do not regenerate a complete animal like plant cells do. Animal body cells are generally not totipotent; only the zygote (and early embryo cells) can form a whole organism.

(d) **Think and mention any two commercial applications of the study above.**

💡 SAMPLE ANSWER

- Growing a large number of identical plants quickly from a small piece (micropropagation), e.g. banana, orchids.
- Producing disease-free plants, saving rare plants, and making improved or genetically modified crops.

⭐ Revise, Reflect, Refine

1 **Meristematic tissues divide repeatedly. What property of their cells allows them to do this?**

(i) Thick walls for protection

(ii) Large vacuoles that store nutrients

(iii) Thin walls, dense cytoplasm and large prominent nucleus

(iv) Functionally differentiated cells

✔ ANSWER

(iii) They have thin walls, dense cytoplasm and a large prominent nucleus, which suit rapid cell division.

2 **If a plant is unable to transport food from leaves to roots, which tissue is malfunctioning?**

(i) Xylem (ii) Phloem (iii) Epidermis (iv) Sclerenchyma

✔ ANSWER

(ii) Phloem: its sieve tubes carry food from leaves to other parts.

3 **Why are the epithelial tissues that line an animal's internal organs usually only one or a few cells thick?**

(i) To store food efficiently (ii) To provide maximum strength (iii) To allow quick exchange of materials across them (iv) To reduce friction

✔ ANSWER

(iii) A thin layer lets substances pass across it quickly (diffusion, absorption).

4 **Do two jumps: straight-leg jump (knees and ankles stiff) and normal jump (knees and**

ankles bend naturally). How did your ankle, knee and hip positions differ between the two jumps?

💡 SAMPLE ANSWER

- Straight-leg jump: ankles, knees and hips stay almost straight; the jump is low and the landing feels hard and jerky.
- Normal jump: ankles, knees and hips bend before take-off and straighten to push the body up; they bend again on landing to absorb the shock. The jump is higher and landing is soft.

So bending at the joints lets muscles work better and protects the joints.

5 **Which type of joint is involved when you bend your knees and ankles?**

(i) Ball and socket (ii) Hinge (iii) Pivot

✔ ANSWER

(ii) Hinge: the knee bends and straightens in one direction, like a door hinge.

6 **Choose: (i) Both A and R true, R explains A; (ii) Both true, R does not explain A; (iii) A true, R false; (iv) A false, R true.**

A. Assertion: Epithelium is well-suited for gas exchange in the lungs. Reason: It consists of multiple layers of tall cells that slow down diffusion.

B. Assertion: Cardiac muscle can contract continuously without fatigue. Reason: Cardiac muscle cells have a high number of mitochondria and an abundant blood supply.

C. Assertion: Tendons connect bone to bone and allow joint movement. Reason: Tendons are made of tough connective tissue that transmits force from muscle to bone.

D. Assertion: In a hinge joint, movement occurs primarily in one plane. Reason: The bone ends are shaped to allow sliding in all directions.

✔ ANSWER

- A: **(iii)**. Lung epithelium is a single layer of thin, flat cells that allows fast diffusion, so the Reason is false.
- B: **(i)**. Many mitochondria and a rich blood supply give constant energy, so the heart beats without tiring.
- C: **(iv)**. Tendons connect muscle to bone (ligaments connect bone to bone); the Reason is true.
- D: **(iii)**. A hinge joint moves in one direction only; its bone ends do not allow sliding in all directions.

7 **Plot a graph of age of a teak tree (x-axis) against diameter (cm) and number of annual rings (y-axis), using Table 3.7.**

Age (years)	DBH (cm)	Annual rings

5	4	5
10	8	10
20	24	20
25	28	25
30	32	30
40	40	40

(i) Analyse the graph for the diameter of the stem over time. (ii) What is the relation between diameter and annual rings? (iii) Which specialised tissue is responsible for the girth of the stem and where is it located?

✔ ANSWER

Graph: plot age on x-axis (0–40 years) and both diameter and rings on y-axis (0–40); join the points to get two lines.

(i) Diameter increases continuously with age. Growth is about 0.8 cm per year from 5 to 10 years, fastest between 10 and 20 years (8 → 24 cm, 1.6 cm per year), and again about 0.8 cm per year after 20 years.

(ii) One annual ring forms each year, so number of rings = age. As rings increase, the diameter also increases; more rings mean a thicker trunk (but the rings are not all of equal width: wide rings in good years, narrow in poor years).

(iii) The **lateral meristem**: a ring of actively dividing cells along the circumference of the stem (between xylem and phloem).

8 **A tree was severely debarked by an elephant (Fig. 3.22).**

(i) Which function(s) of the tree is/are hampered by debarking?

(ii) Which plant tissue would be affected by further damage to the trunk even after debarking?

(iii) Which function would be hampered if the tissues beneath the bark were severely damaged?

(iv) What assumptions are you making? How would the answer change if your assumptions are changed?

💡 **SAMPLE ANSWER**

(i) Protection is lost (more water loss, entry of germs and insects). The phloem just under the bark is removed, so food cannot go from leaves to roots.

(ii) The lateral meristem and then the xylem would be damaged.

(iii) Transport of water and minerals (xylem) and growth in girth (lateral meristem) would stop; the tree may dry up and die.

(iv) Assumption: the bark was removed all around the trunk along with the phloem. If only a small patch or only the outer dead bark was removed, food could still move through the remaining phloem, the wound could heal and the tree would survive.

9 **Aamrapali observed that a young mango sapling's stem bends flexibly during monsoon winds and does not break. Which tissue is responsible for this flexibility? Predict the impact if this tissue was replaced by sclerenchyma.**

✔ ANSWER

Collenchyma: living cells with corners thickened by pectin, which give support with flexibility.

If replaced by sclerenchyma (dead, thick lignified walls), the stem would become hard and rigid. It could not bend in strong wind and might break.

10 **Sohan grew sugarcane from two types of cuttings, A and B (Fig. 3.23). Type B sprouted and grew; type A did not.**

(i) Why could type B grow but not type A? (ii) What difference was present in type B compared to type A? (iii) What observation or measurement was made to see whether this change had an effect? (iv) What parameters should be kept the same for a fair comparison?

💡 SAMPLE ANSWER

(i) Type B had a node, where buds and meristematic (intercalary) tissue are present; these cells divide to form new shoots and roots. Type A had no node, so no new growth.

(ii) Type B had a node (with a bud); type A was only a piece of internode. (check the picture in your book)

(iii) Whether the cuttings sprouted, and the number and length of new shoots and roots after a few weeks.

(iv) Same length and thickness of cuttings, same sugarcane variety, same soil, water, sunlight, temperature, planting depth and time.

11 **Rohan says, "A tissue is a group of similar cells performing similar functions". Rajiv argues, "this is true for simple tissues but a little different for complex tissues". Give your explanation.**

💡 **SAMPLE ANSWER**

Both are partly right. Rohan's statement fits **simple tissues** like parenchyma, collenchyma and sclerenchyma, which have one type of cell.

Rajiv is right about **complex tissues**: xylem (tracheids, vessels, xylem parenchyma, fibres) and phloem (sieve tubes, companion cells, phloem parenchyma, fibres) have different types of cells that work together for one common function (conduction).

So a better definition: a tissue is a group of cells working together to perform a specific function.

12 **Coconut husk fibres are used for mats which are tough and fibrous. Which tissue has structural features suitable for this strength? Explain why living parenchyma couldn't serve the same purpose.**

✔ ANSWER

Sclerenchyma: long, dead cells with thick walls hardened by lignin, so they are hard and strong.

Parenchyma has thin walls, is soft and loosely packed, and being living with cytoplasm it would dry, shrink or rot. So it cannot give strength or last long.

13 **Vibha says, "Meristematic cells are located only at the root and shoot apices". What do you think? What question can Neha ask Vibha to help her understand?**

💡 SAMPLE ANSWER

The statement is **incorrect**. Apical meristems are at the tips, but there are also lateral meristems (along the stem's circumference, for girth) and intercalary meristems (at nodes/base of internodes, for regrowth).

Neha can ask: “If meristems are only at the tips, how does a tree trunk become thicker, and how does grass grow again after it is mowed?”

14 **A plant cell and an animal cell are of the same size.**
(i) Which cell will have a larger vacuole? Give reasons.
(ii) What assumptions are you making?

💡 **SAMPLE ANSWER**

(i) The plant cell. A mature plant cell has a large central vacuole that stores water and substances and keeps the cell firm (turgid). Animal cells have small vacuoles or none.

(ii) Assumptions: both are mature, typical cells; the plant cell is not a meristematic cell (meristematic cells lack vacuoles); the animal cell is not a special cell like a fat-storing cell.

15 **A textbook states, “Each plant tissue performs only one specific function”. What questions would you ask to examine this statement? What examples of tissues would you take?**

💡 **SAMPLE ANSWER**

Questions: Does any tissue do more than one job? Does a storage tissue also make food? Does a conducting tissue also give support?

- Parenchyma: stores food, does photosynthesis in green parts, and forms air spaces for floating.
- Xylem: carries water and minerals and also gives strength.
- Epidermis: protects and, with stomata, helps in gas exchange and transpiration.

So the statement is not fully correct; a tissue has a main function but may do others too.

⭐ The Journey Beyond

1 **Reflect on physical practices you know, such as yoga or kabaddi. How would they support bone and muscle health?**

💡 SAMPLE ANSWER

They make muscles stronger and joints flexible, improve posture and balance, strengthen bones (weight-bearing exercise), and improve breathing and blood flow.

2 **Reflect on gardening methods such as pruning, grafting, irrigation or crop rotation. How does each support plant tissues like meristems, conducting tissues or supporting tissues?**

💡 SAMPLE ANSWER

- Pruning: cutting tips makes meristems at nodes form new branches; the plant becomes bushy.
- Grafting: the vascular tissues (xylem, phloem) of stock and scion join to work as one plant.
- Irrigation: gives water for xylem transport, turgid cells and growth.
- Crop rotation: keeps soil fertile, so meristems get minerals for growth.

⭐ The Quest Continues...

? Will it be possible to obtain a complete animal from an animal cell like plants? If yes, what would be the advantages and challenges?

💡 SAMPLE ANSWER

Not easily; animal body cells are not naturally totipotent. Scientists have done it by cloning (e.g. Dolly the sheep) and with stem cells.

Advantages: saving endangered animals, copying high-yield farm animals, growing tissues for medical use.

Challenges: very low success rate, health problems in clones, high cost and ethical questions.

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**